

Sampling Distributions

Random Sampling

The random variables $x_1, x_2, \dots, x_n$ are a random sample of size 'n' if the

- a) x_i 's are independent random variables
- b) Every x_i has the same probability distribution.

Statistic

A statistic is any function of the observations in a random sample. eg:- mean ($\bar{x}$), variance (s^2), skewness (b_1), kurtosis (b_2).

The probability distribution of a statistic is called the sampling distribution.

Suppose that a random sample of size 'n' is taken from a normal population with mean μ and variance σ^2 , each observation in this sample $x_1, x_2, \dots, x_n$ is a normally and independently distributed random variable with mean μ and variance σ^2 . Then the sample mean $\bar{x}$,

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

$$= \frac{\mu + \mu + \mu + \dots + \mu}{n}$$

$$= \frac{n\mu}{n}$$

$$\bar{x} = \underline{\underline{\mu}}$$

$$\text{Variance } s^2 = \frac{\sigma^2 + \sigma^2 + \dots + \sigma^2}{n^2}$$

$$= \frac{n\sigma^2}{n^2}$$

$$s^2 = \frac{\sigma^2}{n}$$

Central Limit Theorem

If $x_1, x_2, \dots, x_n$ is a random sample of size n taken from a population with mean μ and variance σ^2 and if $\bar{x}$ is the sample mean, the limiting form of the distribution

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \quad \text{as } n \rightarrow \infty$$

$$\sim N(0, 1)$$

or in general

$$Z = \frac{t - E(t)}{SE(t)} \sim N(0, 1) \quad \text{where}$$

$SE(t)$ is the standard error of t .

- Q. An electronic company manufactures resistors that have an average resistance of 100Ω and standard deviation of 10Ω . The distribution of resistance is normal. Find the probability that a random sample of size $n=25$ will have an average resistance less than 95Ω .

$$\text{Given } \mu = 100\Omega, \sigma = 10, n = 25$$

$$P(X < 95) = ?$$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

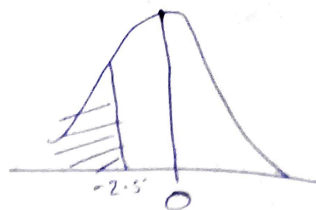
$$\text{When } x = 95, Z = \frac{95 - 100}{10/\sqrt{25}} = \frac{-5}{10} \times 5 = \underline{\underline{-2.5}}$$

$$\text{ie } P(Z < -2.5) = ?$$

$$P(Z < -2.5) = 0.5 - \text{area between } 0 \text{ \& } -2.5$$

$$= 0.5 - 0.4938$$

$$= \underline{\underline{0.0062}}$$



Note

- CLT states that the distribution of sample means approximates a normal distribution as the sample size gets larger, regardless of the population distribution
- Sample sizes equal to or greater than 25 are often considered sufficient for the CLT to hold.
- Large sample size can predict the characteristics of a population more accurately.

... 1. Intervals